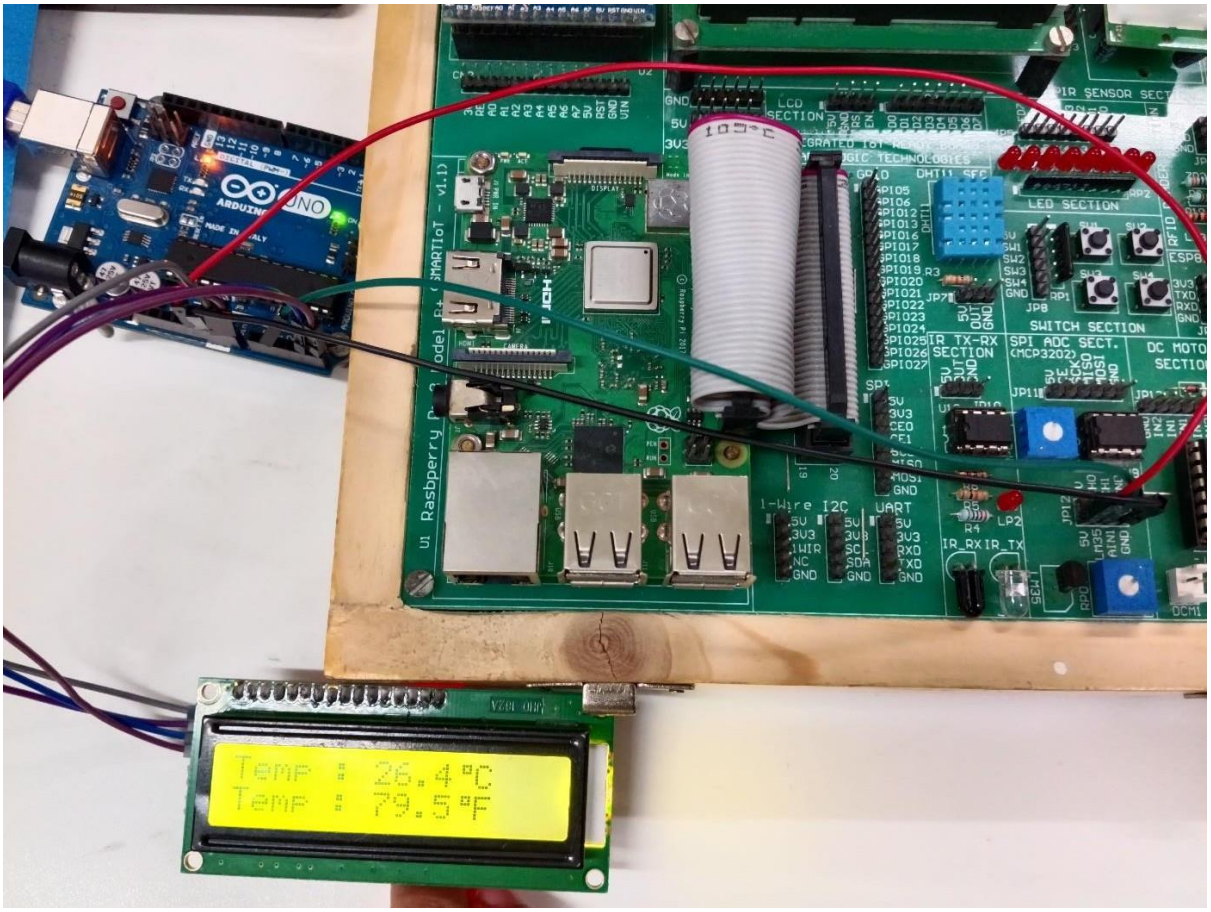
Exp : 3A1 Interfacing LM35 with LCD

Name: SAURABH SHETKAR

PRN No: 22211027

```
#include <LiquidCrystal_I2C.h>   // Include library for LCD
LiquidCrystal_I2C lcd(0x27, 20, 4);    // Initialize LCD (I2C address, columns, rows)

byte Degree[] = {     // Custom degree character
  B00111, B00101, B00111, B00000, B00000, B00000, B00000, B00000
};

void setup() {
  Serial.begin(9600);     // Start serial communication
  lcd.init();                // Initialize the LCD
  lcd.backlight();       // Turn on LCD backlight
  lcd.createChar(0, Degree);    // Load custom degree character
}

void loop() {
  int read = analogRead(A1);         // Read analog value from LM35
  float voltage = read * (5.0 / 1024.0);       // Convert reading to voltage
  float temperatureC = voltage * 100;       // Convert voltage to Celsius

  lcd.setCursor(0, 0);
  lcd.print("Temp : ");
  lcd.setCursor(7, 0);
  lcd.print(temperatureC, 1);        // Display temperature in Celsius
  lcd.write(0);                          // Display degree symbol
  lcd.print("C");

  float temperatureF = (temperatureC * 9.0 / 5.0) + 32.0;       // Convert Celsius to Fahrenheit

  lcd.setCursor(0, 1);
  lcd.print("Temp : ");
  lcd.setCursor(7, 1);
  lcd.print(temperatureF, 1);        // Display temperature in Fahrenheit
  lcd.write(0);
  lcd.print("F");

  delay(1000);  // Wait for 1 second
}
```

Output :



Exp : 3A2 Interfacing LM35 with OLED

Name: SAURABH SHETKAR

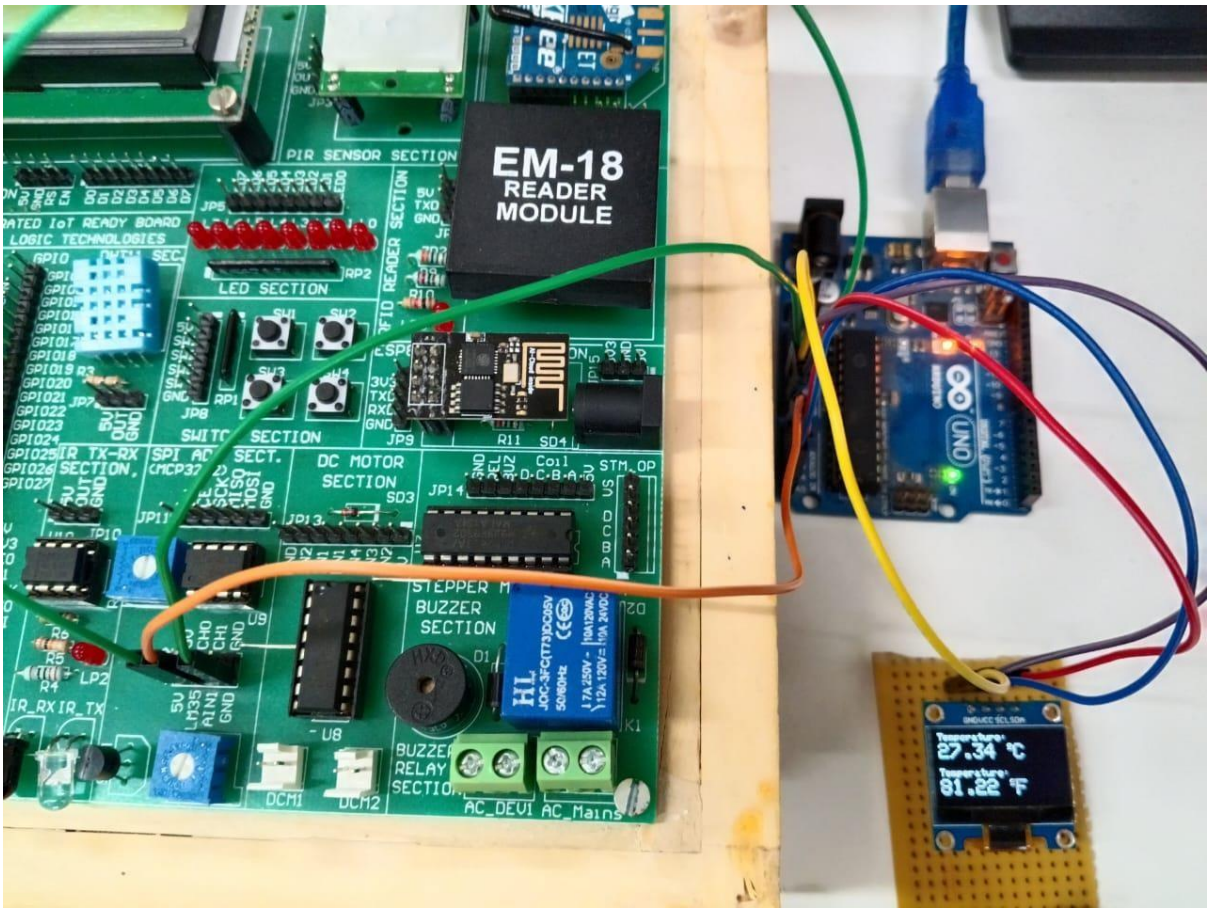
PRN No: 22211027

Code:

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#define SCREEN_WIDTH 128 // OLED display width, in pixels
#define SCREEN_HEIGHT 64 // OLED display height, in pixels

// declaration for an SSD1306 display connected to I2C (SDA, SCL pins)
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, & Wire, -1);
const int sensorPin = A0;
float sensorValue;
float voltageOut;
float temperatureC;
float temperatureF;
// uncomment if using LM35
//float temperatureK;

void setup() {
pinMode(sensorPin, INPUT);
Serial.begin(9600);
if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
Serial.println(F("SSD1306 allocation failed"));

}
delay(2000);
display.clearDisplay();
display.setTextColor(WHITE);
}
void loop() {
sensorValue = analogRead(sensorPin);
voltageOut = sensorValue *( 5.0 / 1024);
// calculate temperature for LM35 (LM35DZ)
temperatureC = voltageOut * 100;
temperatureF = (temperatureC * 1.8) + 32;
// calculate temperature for LM35
//temperatureK = voltageOut / 10;
//temperatureC = temperatureK - 273;
//temperatureF = (temperatureC * 1.8) + 32;
// calculate temperature for LM35
```

```
//temperatureF = voltageOut / 10;
//temperatureC = (temperatureF - 32.0)*(5.0/9.0);
Serial.print("Temperature(oC): ");
Serial.print(temperatureC);
Serial.print(" Temperature(oF): ");
Serial.print(temperatureF);
// clear display
display.clearDisplay();

// display temperature Celsius
display.setTextSize(1);
display.setCursor(0,0);
display.print("Temperature: ");
display.setTextSize(2);
display.setCursor(0,10);
display.print(temperatureC);
display.print(" ");
display.setTextSize(1);
display.cp437(true);
display.write(167);
display.setTextSize(2);
display.print("C");

// display temperature Fahrenheit
display.setTextSize(1);
display.setCursor(0, 35);
display.print("Temperature: ");
display.setTextSize(2);
display.setCursor(0, 45);
display.print(temperatureF);
display.print(" ");
display.setTextSize(1);
display.cp437(true);
display.write(167);
display.setTextSize(2);
display.print("F");
display.display();
delay(1000);
}
```

Output:


EM-18
READER
MODULE
PIR SENSOR SECTION
LED SECTION
DC MOTOR
SECTION
STEPPER M
BUZZER
SECTION
BUZZER
RELAY
SECTION
AC_DEVI
AC_Mains
Temperature:
27.34 °C
Temperature:
81.22 °F

Exp : 3B Interfacing DHT with Arduino

Name: SAURABH SHETKAR

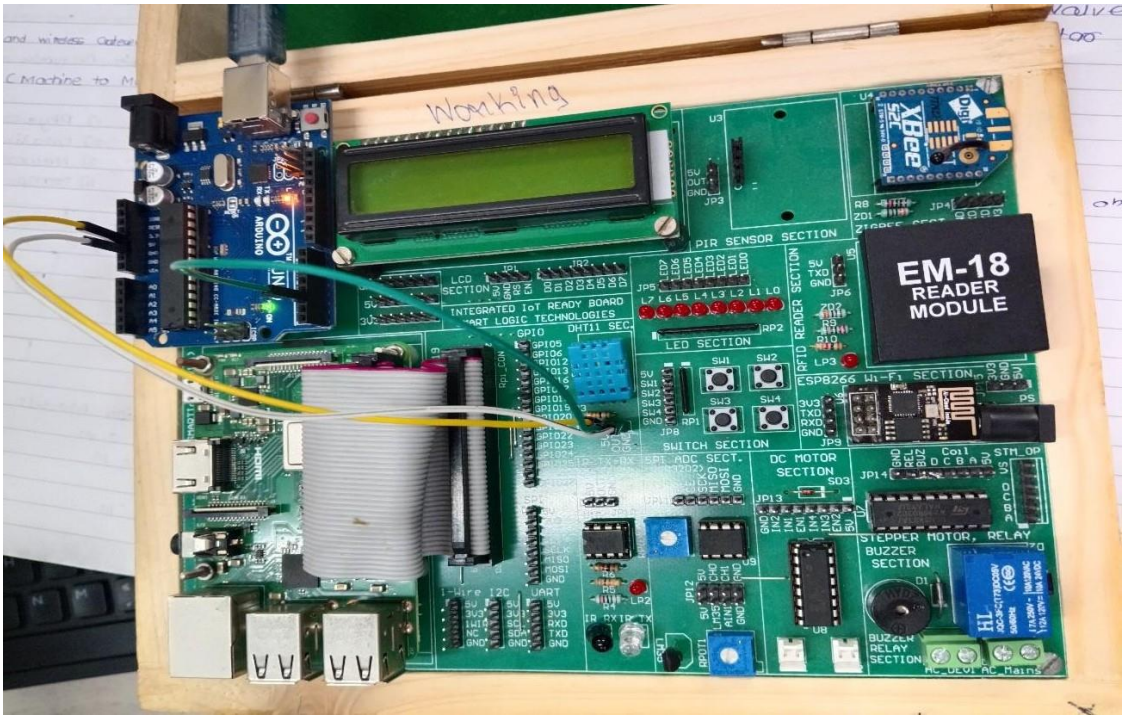
PRN No: 22211027

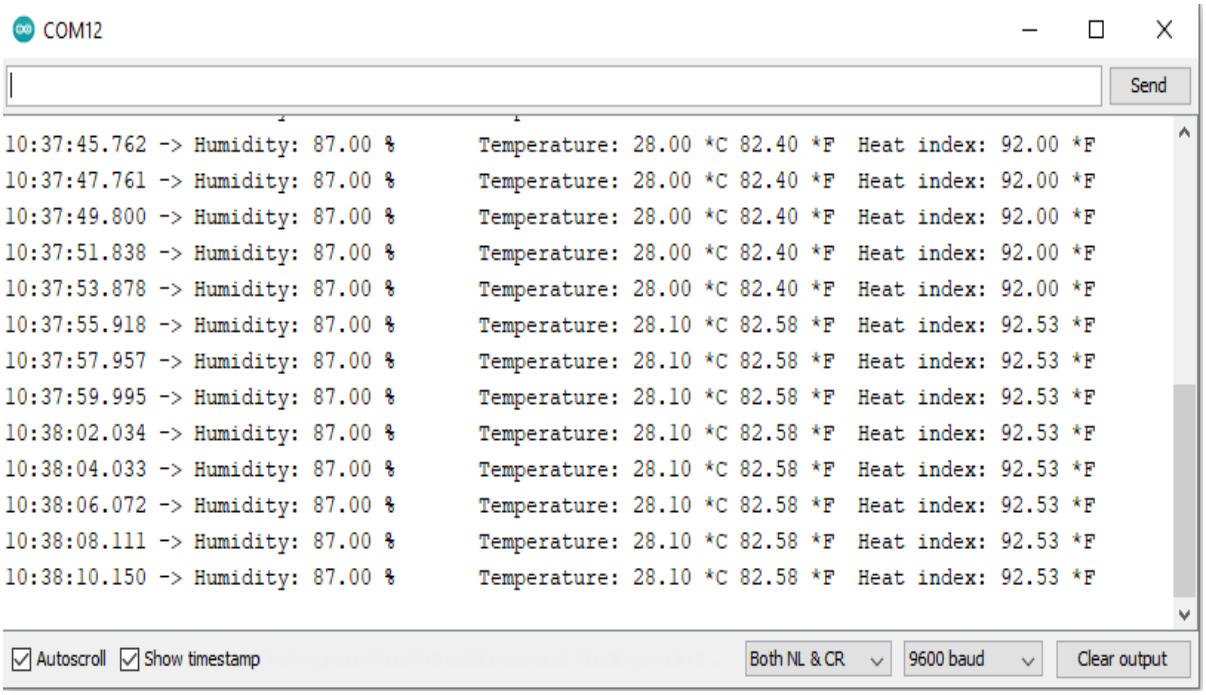
Code:

```
#include "DHT.h"
#define DHTPIN 2 // define the pin for connection
#define DHTTYPE DHT11 // define DHTTYPE DHT 11
DHT dht(DHTPIN, DHTTYPE); // Initialize DHT sensor for normal 16mhz Arduino

void setup() {
Serial.begin(9600);//write a baud rate for serial communication
Serial.println("DHT11 test!");
dht.begin();
}

void loop() {
delay(2000);
// Reading temperature or humidity takes about 250 milliseconds!
// Sensor readings may also be up to 2 seconds 'old' (its a very slow sensor)
float h = dht.readHumidity();
float t = dht.readTemperature();
float f = dht.readTemperature(true);

if (isnan(h) || isnan(t) || isnan(f)) {
Serial.println("Failed to read from DHT sensor!");
return;
}
// Compute heat index
// Must send in temp in Fahrenheit!
float Heat_Index = dht.computeHeatIndex(f, h);//Read Humidity
```

```
Serial.print("Humidity: ");
Serial.print(h);
Serial.print(" %\t");
Serial.print("Temperature: ");// Read temperature as Celsius
Serial.print(t);
Serial.print(" *C ");
Serial.print(f);//Read temperature as Fahrenheit
Serial.print(" *F\t");

Serial.print("Heat index: ");
Serial.print(Heat_Index);
Serial.println("  *F");
}
```

Output:


Working
EM-18
READER
MODULE
PIR SENSOR SECTION
LED SECTION
SWITCH SECTION
DC MOTOR
SECTION
STEPPER MOTOR, RELAY
BUZZER
SECTION
RELAY
SECTION
INTEGRATED IoT READY BOARD
ESP8266 Wi-Fi SECTION


COM12
Send
10:37:45.762 -> Humidity: 87.00 % Temperature: 28.00 *C 82.40 *F Heat index: 92.00 *F
10:37:47.761 -> Humidity: 87.00 % Temperature: 28.00 *C 82.40 *F Heat index: 92.00 *F
10:37:49.800 -> Humidity: 87.00 % Temperature: 28.00 *C 82.40 *F Heat index: 92.00 *F
10:37:51.838 -> Humidity: 87.00 % Temperature: 28.00 *C 82.40 *F Heat index: 92.00 *F
10:37:53.878 -> Humidity: 87.00 % Temperature: 28.00 *C 82.40 *F Heat index: 92.00 *F
10:37:55.918 -> Humidity: 87.00 % Temperature: 28.10 *C 82.58 *F Heat index: 92.53 *F
10:37:57.957 -> Humidity: 87.00 % Temperature: 28.10 *C 82.58 *F Heat index: 92.53 *F
10:37:59.995 -> Humidity: 87.00 % Temperature: 28.10 *C 82.58 *F Heat index: 92.53 *F
10:38:02.034 -> Humidity: 87.00 % Temperature: 28.10 *C 82.58 *F Heat index: 92.53 *F
10:38:04.033 -> Humidity: 87.00 % Temperature: 28.10 *C 82.58 *F Heat index: 92.53 *F
10:38:06.072 -> Humidity: 87.00 % Temperature: 28.10 *C 82.58 *F Heat index: 92.53 *F
10:38:08.111 -> Humidity: 87.00 % Temperature: 28.10 *C 82.58 *F Heat index: 92.53 *F
10:38:10.150 -> Humidity: 87.00 % Temperature: 28.10 *C 82.58 *F Heat index: 92.53 *F
Autoscroll
Show timestamp
Both NL & CR
9600 baud
Clear output

Exp : 3C Interfacing Soil moisture sensor with Relay

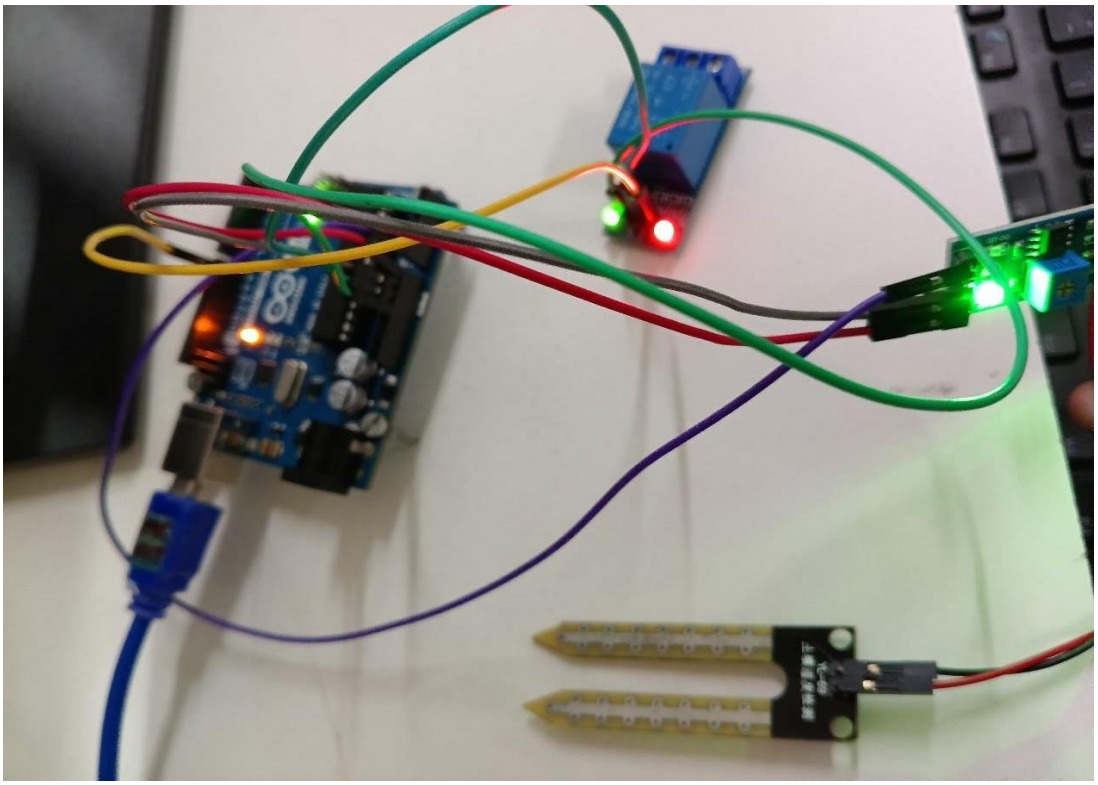
Name: SAURABH SHETKAR

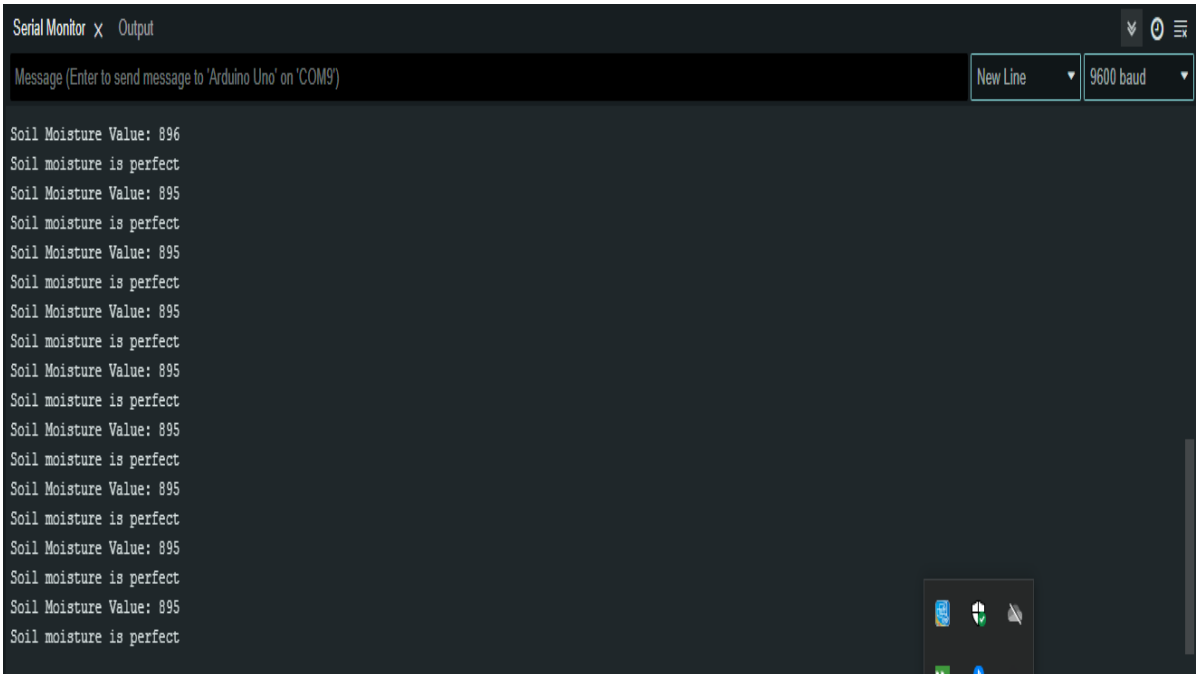
PRN No: 22211027

Code:

```
const int soilMoisturePin = A0;  // Analog pin where soil sensor is connected
const int motorPin = 8;          // Digital pin connected to the relay module
const int moistureThreshold = 500; // Adjust this value based on calibration

void setup() {
  Serial.begin(9600);  // Initialize serial communication for debugging
  pinMode(motorPin, OUTPUT);// Initialize the motor control pin AS Output
  digitalWrite(motorPin, LOW);// Ensure the motor is off
}
void loop() {
  int moistureValue = analogRead(soilMoisturePin);  // Read the soil moisture value from the sensor
  Serial.print("Soil Moisture Value: ");// Print the moisture value to the Serial Monitor
  Serial.println(moistureValue);
  if (moistureValue < moistureThreshold) {
    digitalWrite(motorPin, LOW);// If soil is dry, turn on the motor
    Serial.println("Soil is too dry - time to water!");
  } else {
    digitalWrite(motorPin, HIGH);// If soil is wet enough, turn off the motor
    Serial.println("Soil moisture is perfect");
  }
  delay(5000);
}
```

Output:




Serial Monitor X Output
Message (Enter to send message to 'Arduino Uno' on 'COM9')
New Line
9600 baud
Soil Moisture Value: 896
Soil moisture is perfect
Soil Moisture Value: 895
Soil moisture is perfect
Soil Moisture Value: 895
Soil moisture is perfect
Soil Moisture Value: 895
Soil moisture is perfect
Soil Moisture Value: 895
Soil moisture is perfect
Soil Moisture Value: 895
Soil moisture is perfect
Soil Moisture Value: 895
Soil moisture is perfect
Soil Moisture Value: 895
Soil moisture is perfect
Soil Moisture Value: 895
Soil moisture is perfect

Exp : 3D Interfacing ultrasonic sensor with Arduino

Name: SAURABH SHETKAR

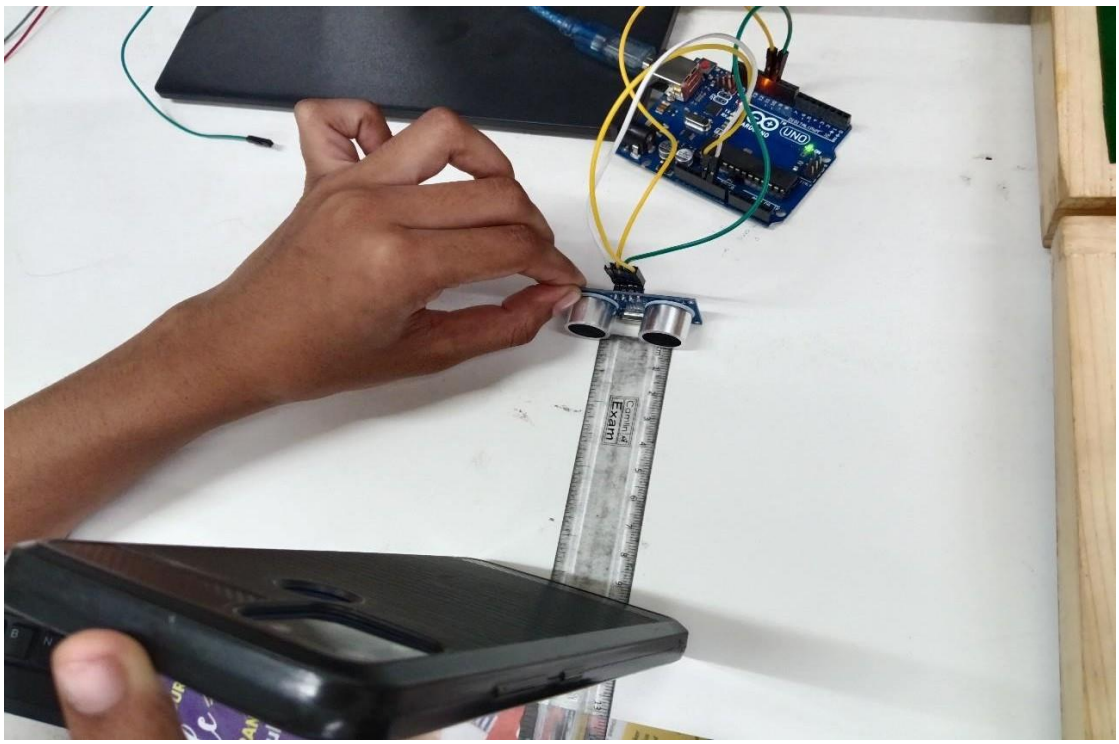
PRN No: 22211027

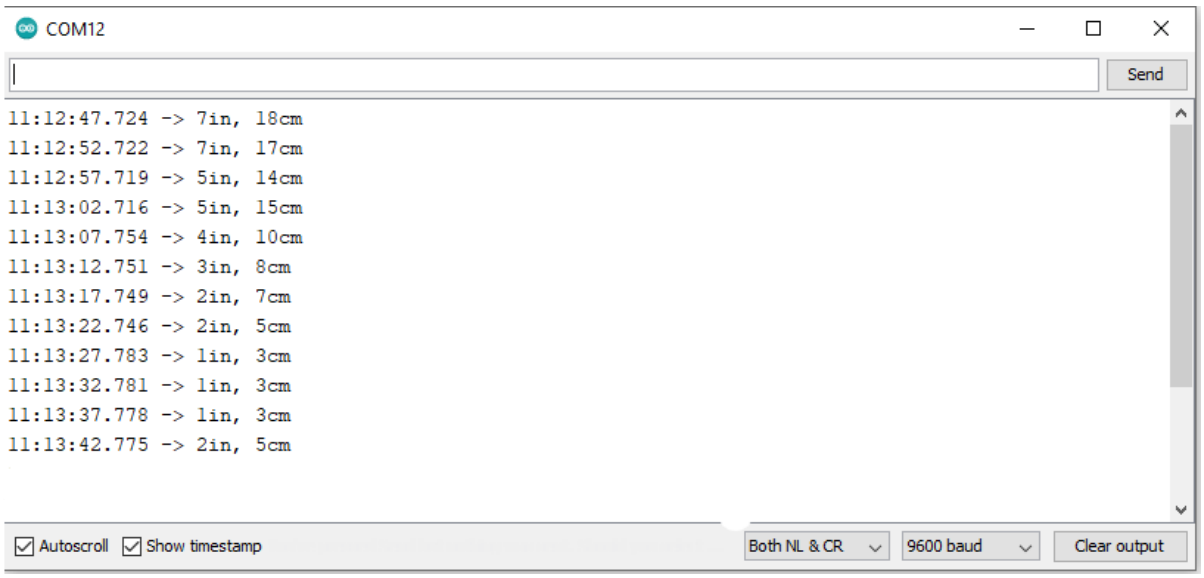
Code:

```
int trigPin = 11; // Define Trigger Pin
int echoPin = 12; // Define Echo Pin
long duration, cm, inches;// Define variable
void setup()
{
Serial.begin (9600);//define pin as off or Ground
pinMode(trigPin, OUTPUT);//define pin as Output pin
pinMode(echoPin, INPUT);//define pin as Input pin
}
void loop() {
digitalWrite(trigPin, LOW);// Give a short LOW pulse beforehand to ensure a clean HIGH pulse:
delayMicroseconds(5);
digitalWrite(trigPin,  HIGH);
delayMicroseconds(10);// duration is the time (in microseconds)
digitalWrite(trigPin, LOW);
// Read the signal from the sensor: a HIGH pulse whose
pinMode(echoPin, INPUT);
duration = pulseIn(echoPin, HIGH);
// Convert the time into a distance
cm = (duration/2) / 29.1; // Divide by 29.1 or multiply by 0.0343
inches = (duration/2) / 74; // Divide by 74 or multiply by 0.0135
Serial.print(inches);
Serial.print("in, ");
Serial.print(cm);
Serial.print("cm");
```

```
Serial.println();

delay(5000);

}
```

Output:





Exp : 3E Interfacing IR sensor module with seven segment IC

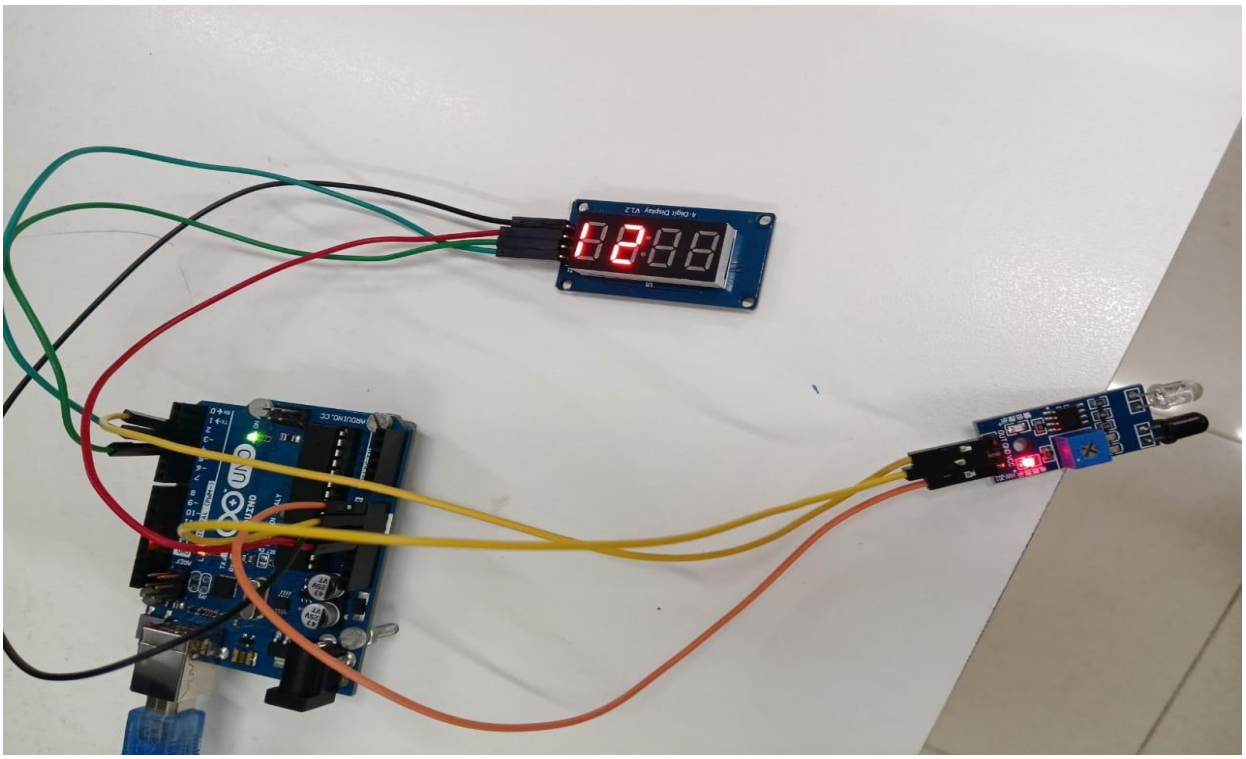
Name: SAURABH SHETKAR

PRN No: 22211027

Code:

```
#include <TM1637Display.h>// Include the library for seven sigment
display

const int irSensorPin = 2; // Digital pin connected to the IR sensor
OUT pin
const int clkPin = 3;      // Digital pin connected to the TM1637 CLK
pin
const int dioPin = 4;      // Digital pin connected to the TM1637 DIO
pin

TM1637Display display(clkPin, dioPin);// Create a display object of
type TM1637Display

int objectCount = 0;//Define a Variable for person measurements
int lastState = LOW;//define lastState as zero
int currentState;//Define a Variable for person measurements

void setup() {
  pinMode(irSensorPin, INPUT); //declare the pin as input
  display.setBrightness(0xff); // Set display brightness to 0x0f
  display.clear();
}

void loop() {
  currentState = digitalRead(irSensorPin);//read the values from the
pin

  // Check if the IR sensor state changed from LOW to HIGH
  if (lastState == LOW && currentState == HIGH) {
    objectCount++;
    display.showNumberDec(objectCount); // Update display with new
count
    delay(200); // Debounce delay
  }

  lastState = currentState;
}
```

Output:



Exp : 3F Interfacing PIR sensor with Arduino

Name: SAURABH SHETKAR

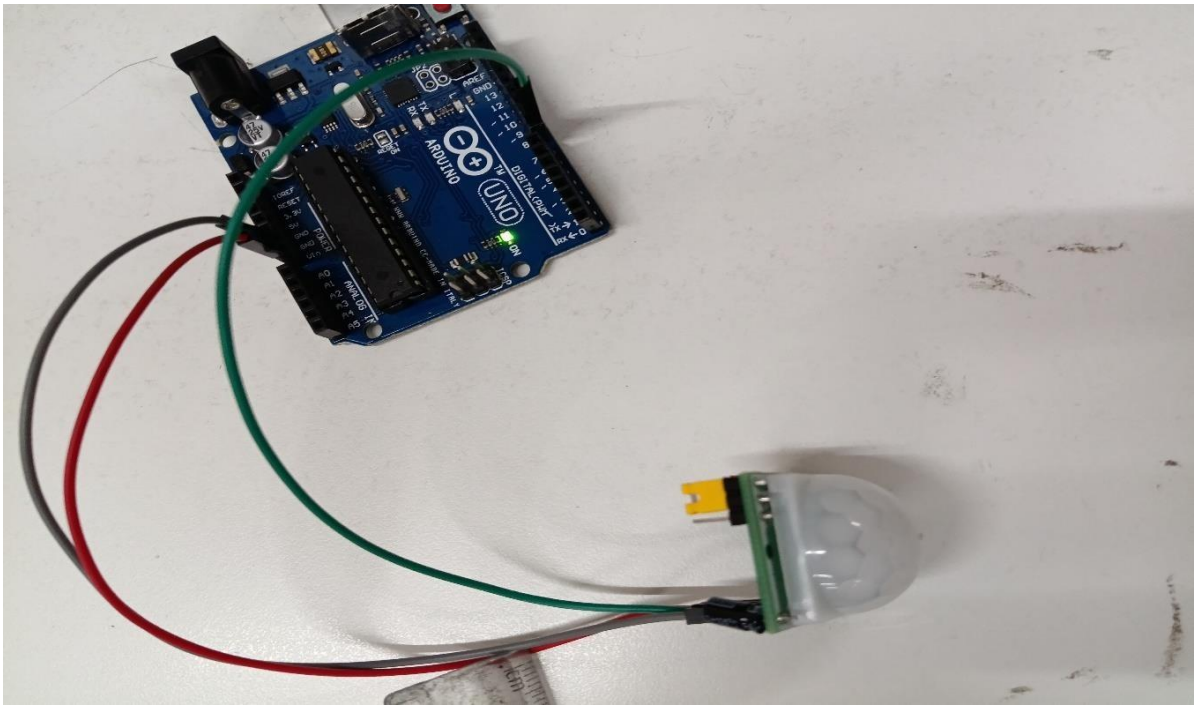
PRN No: 22211027

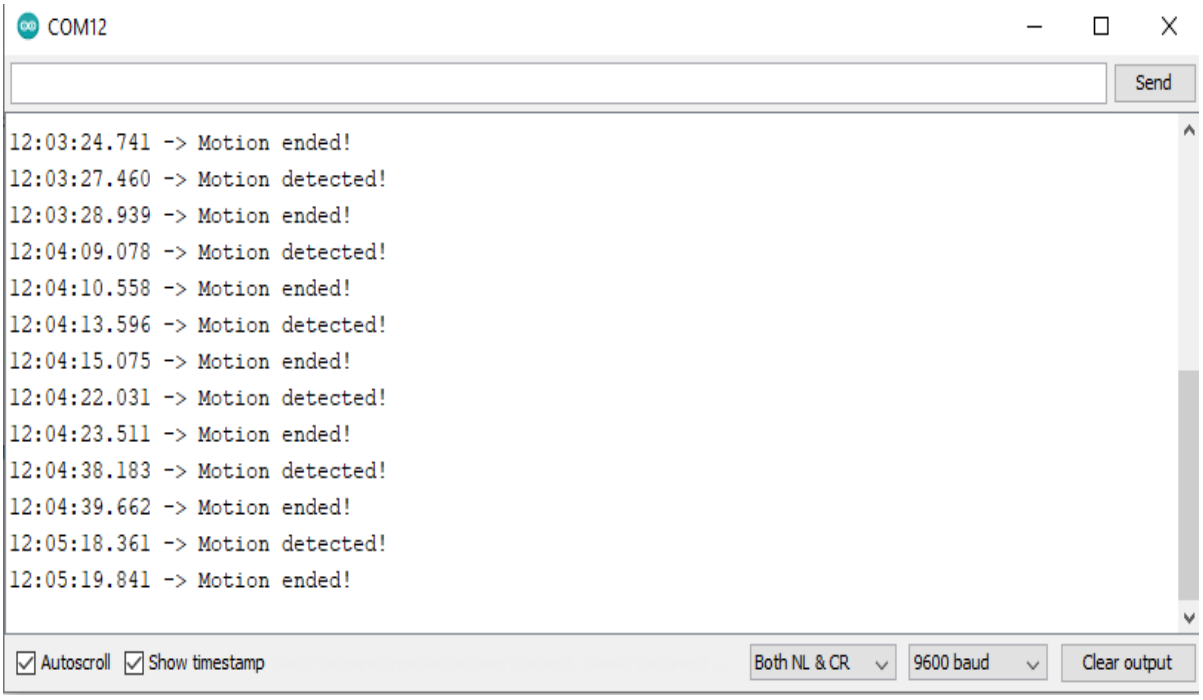
Code:

```
int ledPin = 13; // choose the pin for the LED
int inputPin = 8; // choose the input pin (for PIR sensor)
int pirState = LOW; // we start, assuming no motion detected
int val = 0; // variable for reading the pin status
void setup() {
pinMode(ledPin, OUTPUT); // declare LED as output
pinMode(inputPin, INPUT); // declare sensor as input
Serial.begin(9600);//write a baud rate for serial communication
}
void loop(){
val = digitalRead(inputPin); // read input value
if (val == HIGH) // check if the input is HIGH
{
digitalWrite(ledPin, HIGH); // turn LED ON
if (pirState == LOW)
{
Serial.println("Motion detected!"); // print on output change
pirState = HIGH;
}
}
else
{
digitalWrite(ledPin, LOW); // turn LED OFF
if (pirState == HIGH)
{
Serial.println("Motion ended!"); // print on output change
```

```
pirState = LOW;
}
}
}
```

Output:



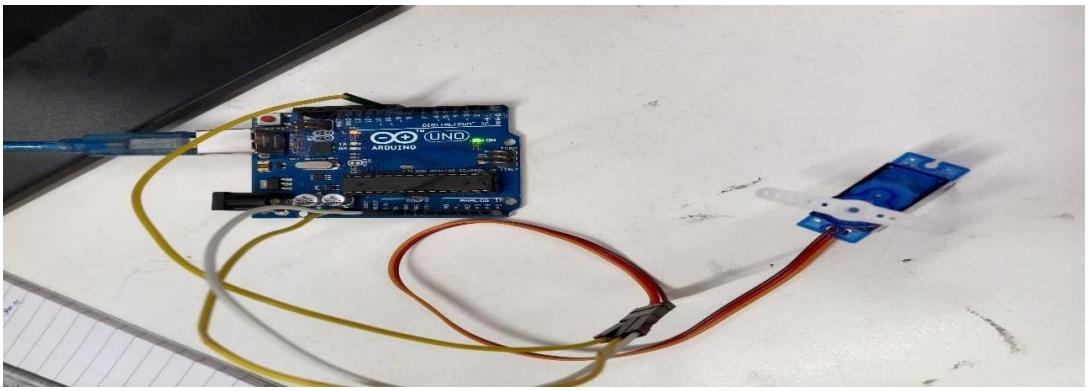


Exp : 3G Interfacing servo motor with Arduino

Name: SAURABH SHETKAR

PRN No: 22211027

Code:

```
#include <Servo.h>//Install the Servo.h library
int servoPin = 9;// choose the input pin (for servo motar)
Servo servo;
int angle = 0; // servo position in degrees
void setup() {
servo.attach(servoPin);
}
void loop() {
// scan from 0 to 180 degrees
for(angle = 0; angle < 180; angle++) {
servo.write(angle);//write the servo angle
delay(15);// write a delay
}
// now scan back from 180 to 0 degrees
for(angle = 180; angle > 0; angle--) {
servo.write(angle);
delay(15);// write a delay
}}
```

Output: